Int. J. Human–Computer Studies 213 (2026) 103824

Contents lists available at ScienceDirect

International Journal of Human - Computer Studies

journal homepage: www.elsevier.com/locate/ijhcs

When AI Disagrees: The Effect of Second Opinion on Patients' Trust in Doctors

Cheng Chen [a,*], Yuan Sun [b], Mengqi Liao [c], S. Shyam Sundar [d,e]

[a] School of Communication, Oregon State University, Corvallis, OR, USA
[b] College of Journalism and Communications, University of Florida, Gainesville, FL, USA
[c] Grady College of Journalism and Mass Communication, University of Georgia, Athens, GA, USA
[d] Bellisario College of Communications, Penn State University, University Park, PA, USA
[e] Department of Immersive Media Engineering, Sungkyunkwan University, Seoul, Republic of Korea

ARTICLE INFO

Keywords:
AI disagreement
perceived recommendation credibility
perceived doctor laziness
perceived medical uncertainty
trust in doctors
LLMs

ABSTRACT

As we increasingly turn to AI tools for second opinions, how might that affect our trust in doctors? We study this in the context of mental health consultation and medical advice, by examining how agreement or disagreement by an AI system influences patients's trust in their doctors. We conducted an experiment (N=135) in which participants interacted with a Large Language Model (LLM)-simulated doctor during a stress management therapy session. At the end of the consultation, the doctor offered to consult an AI assistant for a second opinion. Results showed that AI agreement enhanced perceived recommendation credibility, whereas AI disagreement increased perceptions of medical uncertainty and doctor laziness. Since patients may attribute varying levels of anthropomorphism to an LLM-simulated doctor, when the doctor was perceived as more human-like, AI disagreement increased perceived medical uncertainty and doctor laziness. In contrast, AI agreement enhanced perceived recommendation credibility. These changes in perceptions further influenced patients' cognitive, affective, and behavioral trust in the doctor. We discuss these findings to advance theoretical understanding of agency negotiation within doctor-patient-AI interactions, offer practical suggestions for how doctors can better communicate AI disagreement and agreement to patients, and provide methodological insights for leveraging LLMs for experimental design and message manipulation. Limitations of using LLMs to impersonate human professionals, as well as the limited generalizability of the findings derived from LLM-generated role-play, are also discussed.

1. Introduction

AI has transformed the field of healthcare. It has become increasingly common for medical professionals to consult AI systems for clinical decision support when talking to their patients (Samhammer et al., 2022; D. Wang et al., 2021). When doing so, they are likely to encounter information that confirms or diverges from their recommendations (Kempt et al., 2023; Kempt & Nagel, 2022; Stempniak, 2022; Zhang, Citardi, et al., 2021). Although such agreements or disagreements may not always be overtly communicated to patients, doctors may feel compelled to disclose the second opinion in the interest of shared decision-making in healthcare (McNutt, 2004; Shay & Lafata, 2015). With this in mind, we posed the following question: How does communicating AI's second opinion, either agreement or disagreement, affect patients' trust in doctors?

From a psychological perspective, agreement can reassure patients, thereby enhancing the perceived credibility of the recommendation, especially when this concurrence is provided by an AI system (Detjen et al., 2025). Based on Warranting Theory (Walther & Park, 2002), the machine-ness of an AI medical assistant is likely to confer a higher warranting value to the doctor's assessments as the recommendations could not be manipulated by the doctor. However, one potential drawback of consulting AI and receiving agreement with the doctor's opinion is the risk of perceiving the doctor as lazy. If patients believe that the doctor is relying on an easily accessible AI system rather than more authoritative resources for recommendation verification, they may perceive the doctor as not putting sufficient effort. This perception could undermine trust by implying that the doctor's recommendation lacks the

* Corresponding Author: Cheng Chen, Ph.D., School of Communication, Oregon State University, Corvallis, OR, USA
E-mail addresses: chenc27@oregonstate.edu (C. Chen), yuan.sun@ufl.edu (Y. Sun), mengqi.liao@uga.edu (M. Liao), sss12@psu.edu (S.S. Sundar).

https://doi.org/10.1016/j.ijhcs.2026.103824
Received 16 December 2024; Received in revised form 8 April 2026; Accepted 20 April 2026
Available online 21 April 2026
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anticipated level of expertise (Madsen, 2018).

Compared to agreement, patients may perceive AI disagreement as confusing, increasing uncertainty during medical consultations. Research on medical decision-making has shown that AI-doctor disagreements can reduce patients' confidence in treatment choices and increase anxiety about medical outcomes (Kempt et al., 2023). Furthermore, patients may perceive the doctor as losing their professional autonomy when being contradicted by an AI assistant, as the doctor was initially granted the freedom to make independent clinical judgments without the influence of external factors, including the AI's recommendation (Salvatore et al., 2018).

We extend the line of research on user perceptions of human-AI collaboration (see, for examples, Chen et al., 2024; Meng et al., 2025; Molina & Sundar, 2022) by introducing two distinctive outcomes of such collaboration, agreement and disagreement, and incorporating them into patient consultations. In this pre-registered experiment, we examined how disclosing an AI's second opinion, whether it aligns with or diverges from a doctor's initial recommendation, affects patients' trust in a doctor. Moreover, we sought to unpack the effect of AI disagreement vs. agreement on patient trust by including four potential mediators: perceived recommendation credibility, perceived doctor laziness, perceived medical uncertainty, and perceived professional autonomy.

Given that LLMs and generative AI tools such as ChatGPT can play multiple conversational roles in human society (Maurya, 2024; Milovic et al., 2024) and have demonstrated effectiveness in providing mental health support using Cognitive Behavioral Therapy (CBT) techniques (Alanezi, 2024), we prompted an LLM-powered chatbot to act as a human doctor who interacted with participants to help them manage stress. At the end of the consultation, the LLM-simulated doctor offered to consult an AI assistant for a second opinion. In this study, we make use of LLMs to create experimental conditions involving doctor-patient communications and generate messages that reflect the intended manipulation of AI agreement vs. disagreement. This allows us to reflect on the applicability of LLMs for this purpose.

The study makes significant theoretical contributions to the growing body of research on users' perceptions of human-AI collaboration. Specifically, it broadens the spectrum of human-AI collaboration by including both agreement and disagreement conditions and examining their distinct effects on patients' trust in doctors. Second, it investigates psychological mechanisms that drive the effect of AI disagreement and agreement on patients' trust in doctors, thus revealing the reasons behind changes in trust in doctors after receiving a second opinion from AI. Additionally, this study demonstrates the potential of using LLMs to create experimental conditions and manipulate intended messages, identifying both the capabilities and limitations of LLMs acting as human doctors in mental health consultations.

2. Literature Review

2.1. Clinical Decision Support Systems (CDSSs) and AI-DSS

Modern medical consultations are often supported by Clinical Decision Support Systems (CDSSs). These systems record the patients' medical information and provide patient-specific assessments or recommendations to clinicians for decision-making (Sim et al., 2001; Sutton et al., 2020). Depending on how the system is built, CDSSs can be broadly classified as knowledge-based vs. non-knowledge-based systems.

The knowledge-based CDSSs have rules that govern their decision-making, and these rules are primarily derived from the medical literature, doctor's clinical practice, and patient inputs (Sim et al., 2001; Sutton et al., 2020). By contrast, non-knowledge-based CDSSs, also known as AI-DSS, use machine learning to identify statistical patterns in training data and assist doctors in decision-making (Sim et al., 2001; Sutton et al., 2020). While AI systems are automated machines capable of gathering evidence and forming diagnoses on their own, the final decisions about treatment remain in human doctors' hands (Topol, 2019). Research shows that AI-DSSs have increased human doctors' confidence and performance (Fogliato et al., 2022; Kaur et al., 2020; Wanner, 2021) and reduced clinical errors (Calisto et al., 2022).

While previous studies have examined physicians' perceptions of AI-DSS (Oh et al., 2019) and AI-aided diagnostic advice (Gaube et al., 2021; Goh et al., 2024), few studies have explored patients' perceptions of AI-assisted decision-making in medical encounters. A review of the literature on perceived human-AI collaboration in the context of healthcare, automated journalism, and content moderation reveals three camps of thoughts.

2.2. Perceptions of Human-AI Collaboration

The first camp emphasizes the positive perceptions of collaboration between humans and AI systems in various domains, such as content moderation (Molina & Sundar, 2022) and news writing (Tandoc Jr et al., 2020; Waddell, 2019). In the context of healthcare, Tobia et al. (2021) found that patients perceive physicians accepting an AI's advice as more reasonable than rejecting it. Likewise, Kempt and Nagel (2022) revealed that consulting an AI's opinion on medical decisions is perceived as more responsible.

The second camp argues that patients may have reservations about human-AI collaboration. For example, 60% of Americans reported discomfort with healthcare providers relying on AI in their care (Tyson et al., 2023). Similarly, when AI helps radiologists interpret X-ray results, patients are less confident in the recommendation (York et al., 2020). In the context of automated journalism, a meta-analysis found that machine authorship has a small negative effect on credibility perceptions (Wang & Huang, 2024). Likewise, a study on AI-mediated communication revealed that users perceived the Airbnb host as less trustworthy when they knew or suspected that the host's profile was generated by an AI (Jakesch et al., 2019).

The literature also identifies a third camp in which patients perceive no difference between a human doctor working alone and one assisted by an AI system in decision-making. For example, Chen et al. (2021) found that patients do not perceive any difference in privacy intrusiveness and patient compliance between an individuating human doctor and an individuating AI-assisted human doctor. A recent study confirms that patients generally do not notice or care about AI assistance in medical recommendations, as long as a human doctor is involved in the diagnosis or treatment (Chen et al., 2024).

Given that most studies have relied on a text-based description to illustrate the collaboration between the doctor and the AI system (see examples, Chen et al., 2024; Chen et al., 2021), the oversimplified operationalization of human-AI collaboration may have resulted in the mixed findings of users' perceptions of such collaboration. Without clear details on how humans and AI collaborate, users may struggle to form consistent evaluations and perceptions. To reconcile the mixed findings in the literature, this study uncovers the human-AI collaboration process by examining the psychological effect of obtaining a second opinion from AI, either agreement or disagreement, on patients' trust in their doctors.

2.3. AI Agreement and Disagreement

Ideally, patients may expect to hear agreement between the doctor and the AI system. However, a study shows that doctors often disregard recommendations from AI systems, ignoring up to 95% of its alerts and suggestions (Moxey et al., 2010). Other studies reveal several reasons behind this rejection, including confirmation bias (Fogliato et al., 2022), concerns about legal liability (Kempt et al., 2023; Tobia et al., 2021), the fear of losing face (Liang & Xue, 2022), and the lack of explanation for the AI's disagreement (Samhammer et al., 2022). We define AI disagreement as the situation where the AI system's recommendation diverges from the doctor's judgment. In contrast, agreement refers to the

situation where both the AI and the doctor reach the same or highly similar conclusions. Given that AI is an automated system capable of autonomous decision-making, we argue that both AI agreement and disagreement with the doctor's initial diagnosis can influence patients' trust.

2.4. Trust in Doctors

Trust is fundamental to building a positive patient-doctor relationship (Hall et al., 2001). Informed by previous research, we define trust in doctors as the degree to which patients believe the doctor will care for their interests in vulnerable and uncertain situations. As a multidimensional construct, trust can have cognitive, affective, and behavioral dimensions (Johnson & Grayson, 2005; Soh et al., 2009).

Cognitive trust is defined as the extent to which patients are willing to rely on the doctor for decision-making based on the doctor's competence and integrity (Johnson & Grayson, 2005). This perception is formed through a rational approach that involves an understanding of medical professionals' training, practices, and experiences.

Aside from cognitive trust, Hall et al. (2001) noted that "the emotional, nonrational component of trust is especially prominent in a medical context" (p. 617). Driven by emotions, affective trust is defined as the emotional bond between doctors and their patients (Edell & Burke, 1987; Soh et al., 2009), which likely forms when the doctor shows care and concern for their patients (Johnson & Grayson, 2005).

As a result of cognitive and affective trust, behavioral trust is likely to develop. Behavioral trust is the patients' willingness to follow the doctor's recommendation or evaluation (Albrecht & Travaglione, 2003; Soh et al., 2009). Like cognitive trust, it is based on effortful evaluation as patients choose to act on the doctor's advice.

We predict that AI disagreement and agreement with the doctor's recommendation are likely to have distinct effects on patients' trust in doctors through different mechanisms, including perceived doctor laziness, perceived recommendation credibility, perceived medical uncertainty, and perceived professional autonomy of the doctor. We introduce each of these potential mediators below.

2.5. Perceived Doctor Laziness

One concern related to the application of AI systems in decision-making is the potential for these systems to signal laziness on the part of medical professionals. This concern is expressed by both healthcare professionals (Asai et al., 2019; Samhammer et al., 2022; Subaveerapandiyan et al., 2023; Van Cauwenberge et al., 2022) and patients (Temple et al., 2023).

There are three factors that contribute to the conceptualization of laziness: an individual's performance, abilities, and the motivation and/or effort applied in a given task (Madsen, 2018). Based on these factors, we define perceived laziness as the extent to which patients believe that a doctor's performance falls short of their abilities due to a lack of effort.

Two relevant theories and concepts may help predict the formation of perceived doctor laziness. One is Expectancy Violations Theory, which suggests that the mismatch between expectations and reality can influence users' evaluations and perceptions (Burgoon, 1993, 2015). Compared to situations where expectations are simply confirmed, either positively or negatively, violations of expectations are argued to play a larger role in determining evaluative outcomes (Burgoon, 1993). As Burgoon (1993) noted: "positive violations...are theorized to produce more positive interaction pattern and outcomes than conformity to expectations; negative violation...are theorized to be detrimental, relative to expectancy confirmation" (p. 40). Based on this theory, doctors are more likely to be perceived as lazy if patients hold positive expectations but encounter an unfavorable reality, for example, observing doctors relying on AI to validate their recommendation.

Relatedly, perceptions of laziness may be influenced by mental shortcuts. When a diagnosis is ambiguous, individuals might rely on the effort heuristic (Kruger et al., 2004), i.e., the belief that great effort signals higher quality, to evaluate a doctor's laziness. Given that AI decision-support systems are designed to automate certain cognitive processes in human decision-making, seeking a second opinion from an AI may be viewed as requiring less effort than not using AI at all. The perceived reduction in effort could, in turn, lead to higher perceptions of the doctor's laziness. Such concerns align with those expressed by physicians, who worry that early use of AI in the diagnostic process may make them appear lazy to patients (Samhammer et al., 2022). While workflow integration matters, it remains unclear how AI agreement (vs. disagreement) affects perceptions of a doctor's laziness when the AI is consulted after the doctor has already given an initial recommendation. Thus, we pose the following research question:

RQ1. How does AI agreement (vs. disagreement) influence perceived doctor laziness and further affect patients' trust in the doctor?

2.6. Perceived Recommendation Credibility

A potential benefit of AI agreeing with a doctor's recommendation is that it may enhance the perceived credibility of the recommendation. Aligned with the definition of message credibility as "an individual's judgement of the veracity of the content of communication" (Appelman & Sundar, 2016, p. 63), we define perceived recommendation credibility as the extent to which patients believe that the medical recommendation provided by the doctor is trustworthy and reliable.

A contributing factor to this enhanced credibility assessment might be the invocation of the positive machine heuristic, a belief that machines are more objective and neutral and therefore, more reliable than humans (Sundar & Kim, 2019). Since the validation comes from a machine, AI as the communication source is likely to activate the positive machine heuristic, leading to higher perceived credibility of the recommendation. AI systems typically operate independently and are resistant to human manipulation. The objective nature of AI systems could lend a higher warranting value to the agreement, according to Warranting Theory (Walther & Park, 2002). The increased warranting value is likely to enhance perceived recommendation credibility. Based on this rationale, we propose the following hypothesis:

H1. AI agreement (vs. disagreement) will increase perceived recommendation credibility, which, in turn, will be associated with higher patient trust in the doctor.

2.7. Perceived Medical Uncertainty

Different from agreement, AI disagreement can result in distinct psychological outcomes. One of them is perceived medical uncertainty. Han (2021) defines uncertainty as a subjective perception of ignorance. As a metacognitive conception, experiencing uncertainty suggests that individuals are aware of the lack of information or knowledge necessary to understand the situation at hand (Nissen & Meuter, 2023). Mishel's (1988) theory of uncertainty in illness suggests that a medical event may be perceived as uncertain when a patient is unable to assign definite meaning to the event or accurately predict its outcomes due to a lack of sufficient cues. Following this conceptualization, we define perceived medical uncertainty as the extent to which patients feel they lack sufficient information to fully understand the true nature of their medical situation. It differs from perceived credibility of the recommendation as it focuses on the overall evaluation of the medical condition rather than the specific recommendation provided by the doctor.

While individuals can experience personal growth as an outcome of uncertainty as it opens up the opportunities for a positive outcome (Clarke-Steffen, 1993; Rosenthal et al., 2001), prior studies have linked medical uncertainty to various negative outcomes, including a reduced sense of control (McKoane & Sherman, 2023) and self-doubt (Sloan et al., 2020). It is also associated with decreased confidence in both the medical profession and oneself (Sloan et al., 2020), as well as less

favorable attitudes toward doctor-patient communication (Gala et al., 2022).

Notably, previous research on uncertainty management suggests that credible authorities, such as healthcare providers, can influence the uncertainty that patients experience with their knowledge and communication behaviors (Brashers & Hsieh, 2006). Considering that AI has become a new source of communication, and it might be perceived as credible, if certain positive stereotypes of AI as a machine are triggered, such as the positive machine heuristic (Sundar & Kim, 2019), the disagreement in opinion between two credible sources likely causes confusion about one's medical situation, leading us to hypothesize that AI disagreement will increase perceived medical uncertainty, which in turn would lower patient trust.

H2. AI disagreement (vs. agreement) will increase perceived medical uncertainty, which, in turn, will be associated with lower patient trust in the doctor.

2.8. Perceived Professional Autonomy of the Doctor

AI disagreement may also influence perceived professional autonomy of the doctor. In the clinical context, professional autonomy refers to a doctor's ability to make independent clinical decisions without constraints from external factors such as organizational procedures and financial concerns (Salvatore et al., 2018). However, the applications of AI in various domains of healthcare have significantly impacted healthcare professionals' job autonomy and control (Tursunbayeva & Renkema, 2023). An interview study revealed that radiologists perceived AI as a threat to their professional and epistemic authority (Lombi & Rossero, 2023). The perceived loss of professional autonomy is identified as a primary reason for healthcare professionals' resistance to AI (Lambert et al., 2023).

Previous research also indicates that disputing or rejecting AI's recommendations places the burden of proof on doctors (Kempt et al., 2023), making them vulnerable to conflicts such as medical malpractice claims (Stempniak, 2022). Additionally, physicians may feel uncertain about their actions (Kempt et al., 2023) and compelled to justify their decisions in the case of AI disagreement. Given that clinical reasoning, particularly diagnostic reasoning, forms the core of professional autonomy (Higgs & Jones, cited in Kim & Lee, 2018), and AI systems increasingly take over cognitive activities that were once performed by doctors (Kempt et al., 2023), disagreement from the AI might be seen as a challenge to the doctors' professional autonomy. When patients perceive that the doctors have lost their authority or independence in making final medical decisions, their trust in the doctor may decrease. Based on this rationale, we propose the following hypothesis:

H3. AI disagreement (vs. agreement) will reduce perceived professional autonomy of doctors, which, in turn, will be associated with lower patient trust in the doctor.

2.9. Belief in Machine Heuristic

While AI disagreement is hypothesized to increase perceived medical uncertainty and reduce perceived professional autonomy of the doctor, the effect may depend on patients' prior beliefs or experiences with AI systems. These beliefs or experiences may cohere to form a cognitive heuristic or mental shortcut that guides trust evaluations under uncertainty (Tversky & Kahneman, 1974). A commonly activated heuristic in human-machine interactions is the positive machine heuristic, defined as the extent to which individuals believe that machines are more objective, accurate, and ideologically neutral than humans (Sundar, 2008; Sundar & Kim, 2019; Yang & Sundar, 2024). Previous research shows that individuals with stronger belief in the positive machine heuristic are more likely to agree with the AI's opinion in fact-checking contexts (Banas et al., 2022). Consistent with this evidence, we predict that patients with stronger beliefs in machine heuristic will experience higher perceived medical uncertainty and lower perceived professional autonomy of the doctor during AI disagreement, leading to lower trust in doctors. Thus, we propose the following hypotheses:

H4. The effect of AI disagreement (over agreement) on perceived medical uncertainty will be moderated by patients' belief in the positive machine heuristic, such that for individuals with stronger beliefs in the positive machine heuristic, AI disagreement will increase perceived medical uncertainty, which will, in turn, be associated with lower patient trust in the doctor.

H5. The effect of AI disagreement (over agreement) on perceived professional autonomy of the doctor will be moderated by patients' belief in the positive machine heuristic, such that for individuals with stronger beliefs in the positive machine heuristic, AI disagreement will decrease perceived professional autonomy of the doctor, which will, in turn, be associated with lower patient trust in doctor.

2.10. AI Chatbot in Mental Health Support

Beyond human doctors, previous research suggests that AI chatbots can also provide mental health support, such as delivering supportive and empathetic messages (Meng & Dai, 2021) and employing interpersonal communication strategies, such as individuation (Chen et al., 2024), to improve patient-centered outcomes. In medical contexts where communication is critical, generative AI systems such as ChatGPT have been recognized as effective tools for helping patients restructure their cognition and develop more positive attitudes toward their problem (Alanezi, 2024). Moreover, patients who received psychiatric inpatient care from ChatGPT reported higher quality of life and user satisfaction (Melo et al., 2024). In addition, ChatGPT has demonstrated the ability to provide medically accurate responses comparable to those of professional doctors across multiple medical domains (Girton et al., 2024; Padovan et al., 2024). In the context of using AI for peer support, Young et al. (2024) found that young people preferred AI-generated responses for topics involving relationships, self-expression, and physical health, whereas they preferred human-generated responses for sensitive topics such as suicidal thoughts. Perceptually, patients do not seem to distinguish between human doctors and AI chatbots, especially when the chatbot delivers contingent responses (Meng et al., 2023).

That said, prior studies have highlighted the risks of mental health support delivered by ChatGPT and other LLMs. Some concerns include inappropriate or even harmful recommendations (Dergaa et al., 2024), stigmatization of individuals with mental health conditions, and encouragement of clients' delusional thinking (Moore et al., 2025). In particular, Iftikhar et al. (2025) conducted an 18-month ethnographic study and found that LLM-based counseling frequently violated professional ethical guidelines, including failures in contextual understanding, therapeutic collaboration, authentic empathy, cultural fairness, and crisis safety. These issues reflect the inherent limitations of LLMs, particularly considering their tendency to generate fabricated information (Kozlakidis et al., 2026). This is especially concerning in healthcare contexts, where some patients may lack the knowledge or skills to discern accurate from inaccurate information. When ChatGPT presents misleading or inaccurate recommendations with high confidence, patients may place unwarranted trust in it, leading to potentially harmful consequences for their health and well-being.

2.11. Using LLMs as Research Tools

In addition to providing mental health consultations, LLMs have been used as research tools to generate materials for surveys and experiments (Behrend & Landers, 2025). We extend this line of work by making use of LLMs to elicit the desired manipulation and intended messages through prompting, specifically, doctor-AI disagreement vs. agreement in our study. Our goal is not to evaluate whether LLMs can or

should be used in doctor-patient communications, although our findings may have implications for such potential applications. Rather, we share our experiences of using LLMs to take on specific roles and serve as a research tool for creating experimental conditions and manipulating messages. Recent studies have shown that LLM-generated advertisements (Van Berlo et al., 2024) and health tweets (Schmälzle and Wilcox, 2022) are indistinguishable from human-generated ones, making them ideally suited for use as media stimuli in experiments. Our study extends this use of LLMs for crafting messages from the domain of mass communication to the more conversational context of interpersonal communication.

We present the study model in Figure 1.

3. Method

To address the proposed hypotheses and research question, we conducted a between-subjects online experiment with two conditions: AI disagreement and agreement. The study was approved by the Institutional Review Board at Elon University[1] and pre-registered on Open Science Foundation (OSF) before data collection. Details of the pre-registration can be found on OSF repository at this link: https://osf.io/aku94/overview.

3.1. Developing AI Chatbots and LLM-simulated Doctor

One novelty of the study is that we used LLMs to create experimental conditions and elicit the intended messages about AI disagreement and agreement. Specifically, we built an LLM-powered chatbot using Landbot (https://landbot.io/), which has access to Open AI's API, enabling us to leverage the GPT-3.5 model to develop our AI chatbots. The decision to use an AI chatbot as a proxy for a human doctor is informed by previous research demonstrating the role-playing capabilities of LLMs (Maurya, 2024; Milovic et al., 2024), while acknowledging that LLMs cannot fully replicate the role of a human doctor (Iftikhar et al., 2025; Moore et al., 2025).

Although having real doctors interacting with patients can increase ecological validity, in a controlled experimental setting like this study, it would be impractical to recruit real human doctors to respond to 260 participants. Doing so would introduce confounding factors, as multiple doctors would likely differ in their clinical practices, communication styles, and interaction patterns. In addition, one advantage of using LLM-powered chatbots over traditional, script-based chatbots is their ability to generate personalized responses. Previous research shows that tailored and individualized messages are generally preferred in healthcare contexts (Chen et al., 2024; Chen et al., 2021).

Unlike building traditional chatbots, which relies on user-defined flows and pre-programmed scripts, LLM-powered chatbots are guided by prompts, i.e., natural language commands that guide AI responses. In our prompt design, we began with a simple role-playing instruction, followed by detailed interaction steps. The prompt concluded with instructions on specific issues to avoid in conversations with participants, based on pre-test results. The full prompt script is available in the Appendix.

The conversation began with the LLM-simulated doctor greeting the patient and introducing itself as Dr. Alex. The doctor then asked the patient about the specific source of their daily-life stress. Following this, the LLM-simulated doctor introduced the cognitive behavioral therapy (CBT) approach and invited the patient to engage in a brief, free therapy session using this method. During the session, the doctor guided the patient in identifying a negative thought related to their stressor and encouraged them to re-evaluate the situation by considering more positive perspectives. The session concluded with the doctor recommending CBT as a suitable approach for the patient.

To elicit the intended manipulation, the LLM-simulated doctor then asked participants if they would like to hear the opinion of his AI assistant, CareBot. If participants agreed, the doctor indicated whether CareBot agreed or disagreed with his recommendation. After communicating either AI agreement or disagreement, the LLM-simulated doctor suggested that the patient consider a full therapy session in the future.

3.2. Pre-testing Prompt Effectiveness

We evaluated the prompt's effectiveness with 41 participants recruited from Cloud Research's Connect. After excluding one response suspected to be AI-generated and 12 incomplete responses, 28 responses remained for analysis. Among these, 82.14% (n=23) followed the expected flow. However, we identified a few issues, including robotic response (n=1) and the absence of CareBot's opinion (n=1). Considering the inherent randomness of LLM outputs and the small scale of the identified issues, we proceeded with the main study.

3.3. Main Study Procedure

In the main study, participants completed three steps: 1) filling out a pre-questionnaire, 2) interacting with a doctor online for mental health consultation, and 3) filling out a post-questionnaire. In the pre-questionnaire, participants were asked to provide their demographic information and indicate their familiarity with visiting doctors online and existing belief in machine heuristic. After this, they were informed that the research team developed a new e-health platform designed to provide accessible resources for the public to have mental health consultation. They were offered the opportunity to have a free therapy session with a doctor.

We chose mental health consultation as the study context primarily because mental health issues have increased over the past decade (*Where Mental Illness Is Increasing in the US*, 2023), and AI chatbots have been increasingly used to provide mental health support (Bram, 2022; Zhu et al., 2022). While individuals often turn to non-pharmacological interventions, such as yoga, meditation, and exercise to manage stress and mental health challenges (Saeed et al., 2019), a review study has shown that techniques such as Cognitive Behavioral Therapy (CBT) are also effective in addressing general stress (Hofmann et al., 2012). In clinical settings, patients report improved quality of life after CBT (Grensman et al., 2018) and value learning how to restructure their thoughts and adopt a more positive attitude toward their challenges (Alanezi, 2024). Building on this evidence, we offered participants a free therapy session to help them develop new strategies for managing stress.

After the free session, participants were provided with a four-digit code to proceed with the rest of the questionnaire, in which we measured the mediating variables and dependent variables of interest. Upon completing the post-questionnaire, participants were debriefed about the nature of the study. They were told that the "doctor" they had interacted with was an AI-powered chatbot rather than a human, and that this deception was used to ensure a more controlled experiment. We also provided information about available mental health resources in case they felt uncomfortable after participating in the study.

3.4. Power Analysis and Data Collection

An *a-priori* power analysis for a study with 2 experimental groups indicated that at least 128 participants were needed to achieve the power of .80, an alpha error rate of .05, and a medium effect size (f=.25) using the ANCOVA statistical test in the F tests family. In addition, to estimate the sample size for the proposed interaction effects, we conducted a power analysis using the linear regression test in the F-test family. With one tested predictor and three predictors in total, results revealed that a minimum sample size of 55 participants is required to achieve .80 power and .05 alpha error probability, assuming a medium effect size (f^2=.15). We assumed a medium effect size as a rule of thumb,

[1] The first author was previously affiliated with Elon University.

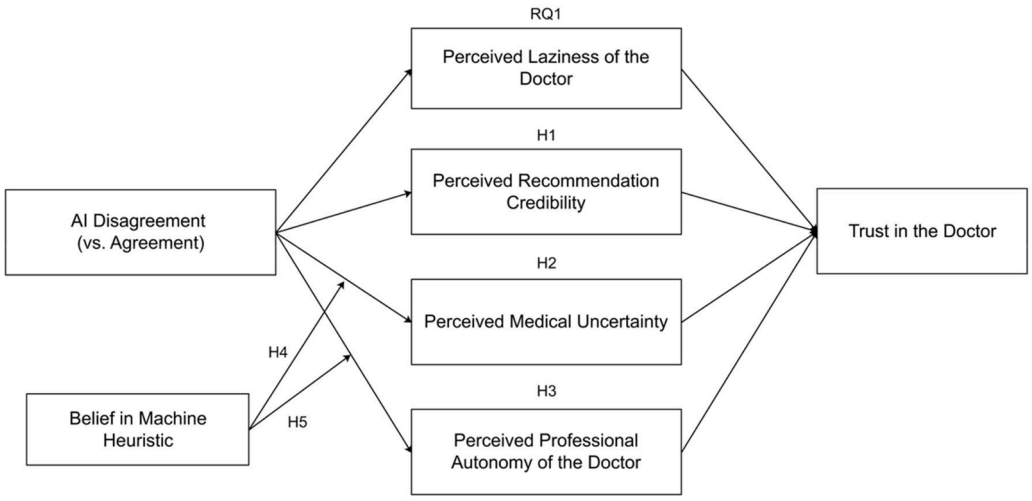


Figure 1. Study model.

given the lack of meta-analysis or similar studies to guide this choice. The implications and limitations of this choice are described in the Discussion section.

To account for potential exclusion of invalid responses due to the inherent variability of LLM outputs, we recruited 260 participants through Research Cloud's Connect platform, using their MTurk panel. Participants who lived in the United States and aged 18 or older were eligible to participate. Based on an hourly rate of $8, participants received $2 as compensation for this 15-min study.

3.5. Testing Prompt Effectiveness

While we initially expected to receive 260 logged responses from Landbot, the platform recorded 264 conversations, likely due to communication issues between Qualtrics and the third-party AI chatbot platform. Since the Landbot logs were not combined with participants' self-reported perceptions from Qualtrics for data analysis, we kept all 264 conversational logs to evaluate prompt effectiveness.

Of these, 31 logs were incomplete due to participants either declining to join the therapy session, not engaging in the therapy session with meaningful and valid responses, failing to answer the LLM-simulated doctor's question, or refusing Carebot's opinion. Among the 233 complete logs, we further excluded 18 cases due to the absence of participants' Connect ID. This occurred either because the doctor chatbot did not request their Connect ID (n=15) or because participants did not report it when prompted by the chatbot doctor (n=3).

Among the remaining 215 logs, 72.09% (n=155) followed the prompts outlined in the Appendix without errors. The most common issues associated with using LLMs for creating experimental conditions included robotic responses (n=18)[2] and instances in which the chatbot failed to understand participants' responses or intentions (n=7). Largely due to the complex set up of the conversational flow, most of the issues happened in the conclusion stage, including not providing the doctor's recommendation (n=16), not asking for participants' preference for CareBot's opinion (n=11), providing CareBot's opinion before asking for participants' preference (n=2), wrong recommendations from the doctor (n=1), and wrong order of the recommendations (n=1). In addition, we observed two instances of hallucination, such as the doctor chatbot making up a source of stress for participants. Consistent with our pretest, we also identified three cases where the doctor chatbot failed to provide participants with the access code at the end of the interaction.

[2] One log showed both a robotic response and the absence of the doctor's recommendation. Thus, it was counted in both categories.

3.6. Participants

Our Qualtrics survey received 260 responses. After excluding those who failed either of the two attention check questions (n=35), those without a matching conversation record on Landbot (n=16), and those whose responses contained errors attributable to the novel use of LLMs for experimental design and message manipulation (n=74), the final sample consisted of 135 participants. The final sample size met the minimum requirement (n=128) for achieving .80 statistical power with a medium effect size (f=.25). Despite the substantial data exclusion, the final dataset retained balanced cell sizes, with 68 participants in the AI agreement condition and 67 in the disagreement condition. The data cleaning process is visualized in Figure 2.

Our final sample consisted of slightly more females (49.6%) than males (48.9%). One participant chose the non-binary option (0.7%) and one preferred not to reveal their gender (0.7%). Their age ranged from 18 to 71 (M=36.49, SD=12.21). The median educational background was a 4-year bachelor's degree, and the median family income was in the range of $50,000 to less than $75,000 a year. Our participants were predominantly White (67.4%), followed by Asian (15.6%), Black or African American (13.3%), Hispanic, Latino, or Spanish origin (8.9%), Native Hawaiian or other Pacific Islander (1.5%), and other race/origin (0.7%).

3.7. Measures

We measured demographics and control variables in the pre-interaction questionnaire. After visiting the chatbot doctor online, we measured mediating and dependent variables in the post-interaction questionnaire. All measures were administered on a 7-point scale.

3.7.1. Mediating Variables

Perceived Recommendation Credibility. Adopted from Appelman and Sundar (2016), participants were asked to indicate their level of agreement with the following statements: The recommendation provided by the doctor is 1) accurate, 2) authentic, and 3) believable. We formed an index of perceived recommendation credibility by averaging the three items, which was reliable (Cronbach's α=.91, M=4.91, SD=1.44).

Perceived Doctor Laziness. Based on the conceptualization of lack of effort, we developed four items to assess perceived doctor laziness, including 1) "The doctor did not show the level of effort that corresponds to his profession;" 2) "The doctor did not live up to the expected expertise;" 3) "The doctor gave a recommendation that easily came to his mind;" and 4) "The doctor was lazy when providing the

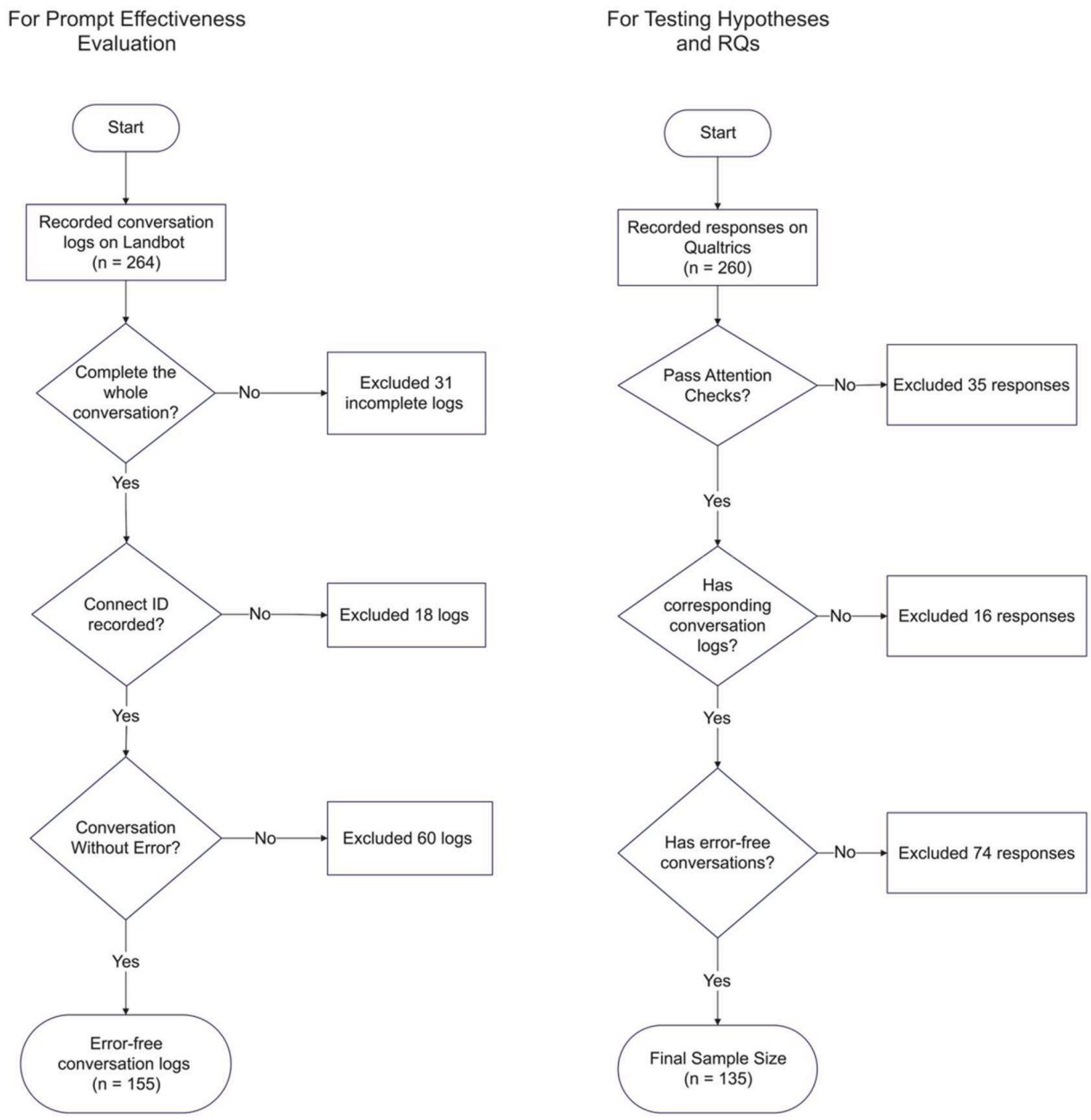


Figure 2. Data cleaning process for conversational logs (left) and Qualtrics survey responses (right).

recommendation." A principal component analysis revealed that item 3 had a loading below .40, leading us to remove it. We then created an index by averaging the remaining three items (Cronbach's α=.86, *M*=4.21, *SD*=1.53).

Perceived Professional Autonomy of the Doctor. We adopted one item from Lin (2014) and developed three similar items to measure perceived professional autonomy. Sample items included: "The doctor was independent in his decision-making" and "The doctor made up his mind by himself." We averaged the four items to form an index of perceived professional autonomy, which had good reliability (Cronbach's α=.90, *M*=4.79, *SD*=1.36).

Perceived Medical Uncertainty. We adopted six items from the Uncertainty of Illness Scale (Lin et al., 2012; Mishel, 1981) to evaluate perceived medical uncertainty. Sample items include: "I was not confident about the decision made by the doctor" and "The doctor said things to me that were confusing." Given that the factor loading of the second item ("I felt the doctor could not provide a better recommendation") was very low compared to the other five items, we excluded it from the measure. We then averaged the rest of the five items to form the index of perceived medical uncertainty, which was reliable (Cronbach's α=.90, *M*=3.87, *SD*=1.33).

3.7.2. Moderating Variable

Belief in Machine Heuristic. Adapted from Yang and Sundar (2024), we used seven items to measure participants' belief in the positive machine heuristic. Sample items include: "AIs are more reliable than humans" and "AIs are more dependable than humans" We averaged the seven items to create an index, which was reliable (Cronbach's α=.95, *M*=3.04, *SD*=1.42).

3.7.3. Dependent Variables

Cognitive Trust. Modified from previous studies (Molina & Sundar, 2022; Ohanian, 1990), we asked participants to rate the doctor on four pairs of adjectives: undependable/dependable, dishonest/honest, insincere/sincere, and untrustworthy/trustworthy. The index of cognitive trust was reliable (Cronbach's α=.94, *M*=4.39, *SD*=1.68).

Affective Trust. We adopted three items from Johnson and Grayson (2005) and developed three similar ones to measure affective trust in the doctor. Sample items include: "The doctor displayed a warm and caring attitude towards me" and "I can talk freely with the doctor about my mental health concerns." Both item 4 ("I would feel a sense of loss if I were not able to see this doctor again") and item 5 ("I feel an emotional bond with the doctor") had low factor loading. After removing the items, the remaining items were averaged to create an index of affective trust (M=3.60, SD=1.50, Cronbach's α=.92).

Behavioral Trust. We modified four items from previous studies (Chen et al., 2021; DiMatteo et al., 1993; Soh et al., 2009) to measure behavioral trust. One example is: "I plan to follow the doctor's recommendation to take care of my mental health." We averaged the four items and created an index of behavioral trust, which had good reliability (Cronbach's α=.97, M=4.02, SD=1.77).

3.7.4. Control Variables

Familiarity with Visiting Doctors Online. We used three items from prior research (Chen et al., 2024) to evaluate participants' familiarity with visiting doctors online. Sample items include: "I have experience visiting doctors online" and "I have visited doctors online in the past." We averaged the three items to form an index (M=4.61, SD=1.90, Cronbach's α=.95).

Perceived Doctor Anthropomorphism. Adopted from Bartneck et al. (2009), participants were asked to indicate their perceptions of the doctor based on four pairs of adjectives, including 1) artificial/natural, 2) synthetic/real, 3) inanimate/living, and 4) machinelike/humanlike. These perceptions were rated on a 7-point scale. We created an index of perceived anthropomorphism, which was reliable (Cronbach's α=.96, M=3.98, SD=2.03). The kurtosis value (-1.2) and skewness value (-.007) both fall within the acceptable range for normality, according to the criteria proposed by Hair et al. (1998). Moreover, a two-tailed independent sample t-test showed that perceived anthropomorphism did not vary significantly between the AI disagreement condition (M=3.96, SD=2.03) and the AI agreement condition (M=4.00, SD=2.00), t (133)=-.10, p=.923, Cohen's d=-.02.

3.8. Construct Validity

To assess convergent and discriminant validity of the key study variables, we followed Fornell and Larcker's (1981) criteria. Convergent validity was achieved if the average variance extracted (AVE) was greater than 0.50 and the composite reliability (CR) was higher than 0.70. Discriminant validity was established if the square root of the AVE for each construct was greater than the correlation coefficients involving that construct. According to these criteria, the key study variables all showed good convergent and discriminant validity, as shown in Table 1.

3.9. Data Analysis Plan

Since our model included three theoretically related dependent variables, namely, cognitive trust, affective trust, and behavioral trust, we used path analysis through SPSS Amos to allow these outcomes to be modeled simultaneously while accounting for their intercorrelations. Although some mediators are correlated, we tested each mediation and moderated mediation model separately because the mediators in our model represent theoretically distinct psychological constructs, and prior literature does not indicate that they should systematically covary with one another. The goal of mediation analysis is to help understand how a causal variable influences an outcome variable through one or more intervening variables, or mediators (Little, cited in Kline, 2015), whereas the moderated mediation analysis, in contrast, identifies the condition(s) under which the mediation effect holds.

In each analysis, we used 2,000 bootstrap resamples and 90% percentile bootstrap confidence intervals (CIs). The analyses were conducted using matrix data after controlling for perceived doctor anthropomorphism. Since familiarity with visiting doctors online did not significantly affect any of the mediating or dependent variables, we did not include it as a covariate.

To account for Type I error inflation resulting from conducting multiple significance tests, we used false discovery rate control, which provides more flexibility when working with emerging theoretical frameworks (Matsunaga, 2007). Specifically, we implemented the Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) by rank-ordering the p-values for the m tests within each family and identifying the threshold using the formula $(\alpha/m) \times i$, where i denotes the rank order and α was set to .05.

4. Results

We first examined the effectiveness of the manipulation and then answered each of the proposed research questions and hypotheses.

4.1. Manipulation Check

We developed four items to examine perceived human-AI disagreement: 1) CareBot did not agree with the doctor's decision, 2) CareBot did not support the decision made by the doctor, 3) CareBot expressed a second opinion differing from the doctor's decision, and 4) CareBot provided an alternative recommendation to the doctor's decision. We formed an index by averaging the four items, which was reliable (Cronbach's α=.92, M=3.56, SD=2.18). A one-tailed independent sample t-test revealed that participants in the disagreement condition (M=5.43, SD=1.24) perceived higher doctor-AI disagreement compared to their counterparts in the agreement condition (M=1.68, SD=1.01), t (98)=16.61, $p < .001$, Cohen's d=3.33. Thus, our manipulation was successful.

4.2. Testing the Mediation Effects

Aligned with the wording in our hypotheses, we used the terms "mediation effects" or "mediation analyses" when presenting the results. Mediation is a form of indirect effect, used to explain how a casual variable influences an outcome through one or more intervening

Table 1
Convergent and Discriminant Validity of Key Study Variables.

Variables	CR	AVE	1	2	3	4	5	6	7	8	9
Credibility (1)	.95	.85	**.92**								
Laziness (2)	.90	.71	-.58***	**.84**							
Autonomy (3)	.93	.83	.30**	-.21*	**.91**						
Uncertainty (4)	.93	.72	-.61***	.70***	-.19	**.85**					
Cognitive Trust (5)	.96	.86	.67***	-.76***	.36***	-.72***	**.93**				
Affective Trust (6)	.94	.72	55***	-.76***	.38***	-.56***	.84***	**.85**			
Behavioral Trust (7)	.98	.93	.66***	-.61***	.43***	-.61***	.75***	.74***	**.96**		
Anthropomorphism (8)	.97	.89	.54***	-.63***	.40***	-.47***	.77***	.79***	.52***	**.94**	
Machine Heuristic (9)	.95	.79	.29**	-.31**	.17*	.13	.39***	.43***	.40***	.35***	**.89**

Note. Bold values on the diagonal are square root of AVE. ***$p < .001$, **$p < .01$, *$p < .05$. $N = 135$.

variables, or mediators (Little, cited in Kline, 2015). Moderated mediation analysis, in contrast, identifies the condition(s) under which the mediation effect holds. In the following analyses, AI disagreement was coded as 1, whereas AI agreement was coded as 0.

4.2.1. Path Analysis with Perceived Doctor Laziness as a Mediator

The path model that included perceived doctor laziness as a mediator demonstrated an acceptable fit to the data, $\chi^2(3)=1.63$, $p=.65$, CFI=1.00, TLI=1.00, RMSEA=.00 (90% CI: .00-.12), SRMR=.02. Results showed that the mediation effect was significant for cognitive trust, B=-.33, SE=.18, 90% CI: [-.64, -.06], $p=.041$, affective trust, B=-.24, SE=.13, 90% CI: [-.48, -.05], $p=.034$, and behavioral trust, B=-.28, SE=.16, 90% CI: [-.57, -.05], $p=.040$. These effects remained significant after controlling the false discovery rate using the Benjamini-Hochberg procedure.

4.2.2. Path Analysis with Perceived Recommendation Credibility as a Mediator

H1 proposed the mediating role of perceived recommendation credibility in the effect of AI agreement on patient trust. Model fit was strong for the path analysis involving perceived recommendation credibility: $\chi^2(3)=.98$, $p=.805$, CFI=1.00, TLI=1.00, RMSEA=.00 (90% CI: .00-.09), SRMR=.02. Results revealed a significant mediation effect on cognitive trust (B=-.31, SE=.15, 90% CI: [-.57, -.08], $p=.023$), affective trust (B=-.12, SE=.08, 90% CI: [-.30, -.03], $p=.023$), and behavioral trust (B=-.33, SE=.16, 90% CI: [-.62, -.09], $p=.023$). The obtained p-values remained significant after applying the Benjamini-Hochberg procedure to control for multiple comparisons. Thus, H1 was supported.

4.2.3. Path Analysis with Perceived Medical Uncertainty as a Mediator

H2 hypothesized perceived medical uncertainty would mediate the effect of AI disagreement on patient trust. The path model that included perceived medical uncertainty exhibited good fit: $\chi^2(3)=1.54$, $p=.67$, CFI=1.00, TLI=.99, RMSEA=.00 (90% CI: .00-.11), SRMR=.02. The mediation effect was significant for cognitive trust (B=-.44, SE=.05, 90% CI: [-.75, -.18], $p=.006$), affective trust (B=-.19, SE=.04, 90% CI: [-.40, -.07], $p=.006$), and behavioral trust (B=-.37, SE=.08, 90% CI: [-.66, -.14], $p=.006$). These effects remained significant after controlling for multiple comparisons using the Benjamini-Hochberg procedure. Thus, H2 was supported.

4.2.4. Path Analysis with Perceived Professional Autonomy as a Mediator

H3 proposed that the effect of AI disagreement on patient trust in the doctor would be mediated by perceived professional autonomy. The model fit of the path analysis incorporating perceived professional autonomy as a mediator was acceptable: $\chi^2(3)=3.41$, $p=.33$, CFI=1.00, TLI=.99, RMSEA=.03 (90% CI: .00-.15), SRMR=.05. However, the mediation effects were not significant for cognitive trust (B=-.02, SE=.05, 90% CI: [-.14, .04], $p=.535$), affective trust (B=-.01, SE=.04, 90% CI: [-.11, .02], $p=.466$), or behavioral trust (B=-.03, SE=.08, 90% CI: [-.20, .08], $p=.596$). Thus, H3 was not supported.

4.3. Testing the Moderated Mediation Models

H4 proposed belief in machine heuristic as a moderator in the mediation effect of AI disagreement on patient trust through perceived medical uncertainty. The path analysis showed good model fit on some, but not all, indicators: $\chi^2(6)=15.54$, $p=.016$, CFI=.98, TLI=.94, RMSEA=.11 (90% CI: .04-.18), SRMR=.04. More importantly, significant moderated mediation effects involving perceived medical uncertainty as a mediator were observed for cognitive trust (B=-.17, SE=.09, 90% CI: [-.32, -.03], $p=.045$), affective trust, (B=-.05, SE=.04, 90% CI: [-.16, -.02], $p=.024$), and behavioral trust, (B=-.12, SE=.07, 90% CI: [-.27, -.03], $p=.042$). These effects remained significant after controlling for false discovery rate using the Benjamini-Hochberg procedure. Thus, H4 was supported. We plot the interaction effect between AI disagreement and belief in machine heuristic on perceived medical uncertainty in Figure 3.

We used a similar approach to examine whether belief in machine heuristic moderated the effect of AI disagreement on patient trust through perceived professional autonomy of the doctor (H5). Model fit of the path analysis was acceptable on some indicators: $\chi^2(6)=10.50$, $p=.105$, CFI=.99, TLI=.97, RMSEA=.08 (90% CI: .00-.15), SRMR=.05. However, the moderated mediation effects were not significant for cognitive trust (B=-.00, SE=.03, 90% CI: [-.05, .05], $p=.680$), affective trust (B=-.00, SE=.02, 90% CI: [-.05, .02], $p=.561$), or behavioral trust (B=-.00, SE=.05, 90% CI: [-.09, .07], $p=.788$). Thus, H5 was not supported.

4.4. Exploratory Data Analysis

We instructed the LLM to role-play as a human doctor; however, it may still have been perceived as an AI when cues reveal its artificial nature. In addition, the level of AI agreement did not significantly affect perceived anthropomorphism of the doctor, $F(1, 133)=.01$, $p=.923$, partial $\eta^2=.00$. This suggests that perceived anthropomorphism was not influenced by our manipulation of AI agreement vs. disagreement. Thus, we explored whether perceived anthropomorphism of the LLM-simulated doctor moderated the effect of AI agreement (vs. disagreement) on patient trust via perceived doctor laziness, perceived medical uncertainty, perceived recommendation credibility, and perceived professional autonomy of the doctor, respectively.

4.4.1. Conditional Indirect Effect of AI Agreement on Patient Trust Through Perceived Doctor Laziness, Moderated by Perceived Doctor Anthropomorphism

We first conducted a path analysis that included perceived anthropomorphism as a moderator and perceived doctor laziness as a mediator. The moderated mediation effects were significant for cognitive trust (B=.35, SE=.08, 90% CI: [.23, .50], $p=.001$), affective trust (B=.30, SE=.07, 90% CI: [.19, .43], $p=.001$), and behavioral trust (B=.31, SE=.08, 90% CI: [.21, .46], $p=.001$). These effects remained significant after controlling for false discovery rate using the Benjamini-Hochberg procedure. As shown in Figure 4, the effect of AI disagreement (vs. agreement) on perceived doctor laziness was stronger among participants who perceived the LLM-simulated doctor as more human-like.

4.4.2. Conditional Indirect Effect of AI Agreement on Patient Trust Through Perceived Recommendation Credibility, Moderated by Perceived Doctor Anthropomorphism

We also tested a model with perceived recommendation credibility as a mediator and perceived anthropomorphism as a moderator. Exploratory analyses showed that the effect of AI agreement on perceived recommendation credibility was moderated by perceived anthropomorphism As shown in Figure 5, AI agreement resulted in higher perceived recommendation credibility when the doctor chatbot was perceived as more human-like. In turn, perceived recommendation credibility was associated with higher cognitive trust (B=.28, SE=.08, 90% CI: [.16, .40], $p=.001$), affective trust (B=.21, SE=.06, 90% CI: [.11, .30], $p=.001$), and behavioral trust (B=.29, SE=.07, 90% CI: [.17, .41], $p=.001$). All effects remained statistically significant after applying the Benjamini-Hochberg procedure to control for multiple comparisons.

4.4.3. Conditional Indirect Effect of AI Disagreement on Patient Trust Through Perceived Medical Uncertainty, Moderated by Perceived Doctor Anthropomorphism

We next examined a model with perceived anthropomorphism as the moderator and perceived medical uncertainty as the mediator. As shown in Figure 6, the effect of AI disagreement on perceived medical uncertainty was conditional on perceived anthropomorphism of the doctor. In turn, perceived medical uncertainty significantly influenced cognitive trust (B=.18, SE=.08, 90% CI: [.08, .33], $p=.006$), affective trust

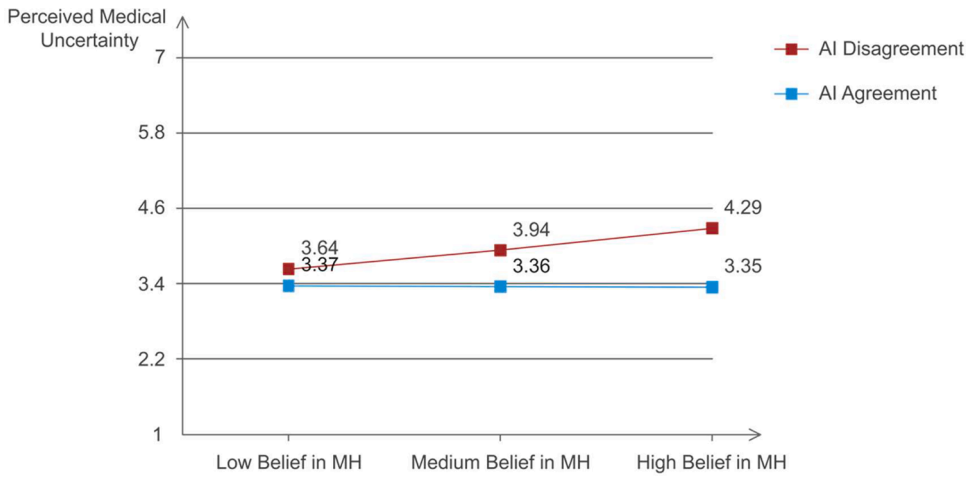


Figure 3. Interaction effect between AI disagreement (vs. agreement) and belief in machine heuristic on perceived medical uncertainty.
Note: MH=Machine Heuristic

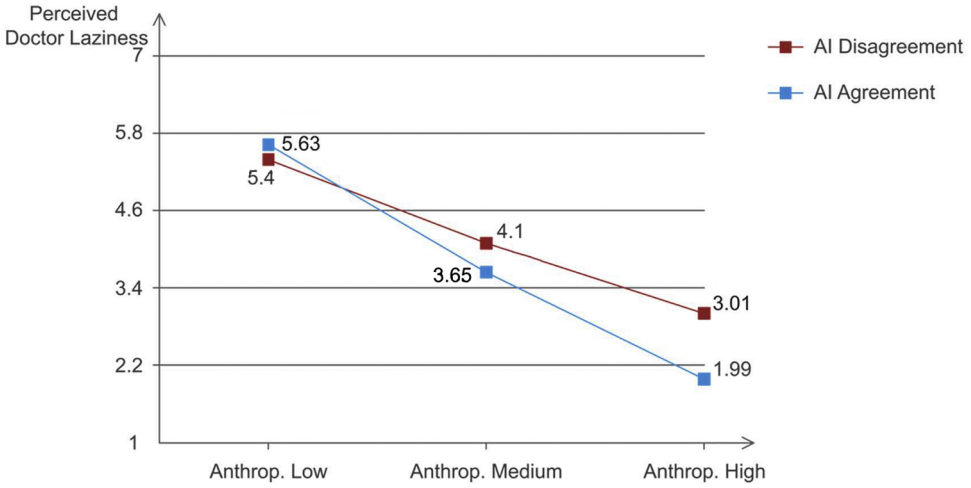


Figure 4. Interaction effect between AI disagreement (vs. agreement) and perceived doctor anthropomorphism on perceived doctor laziness.

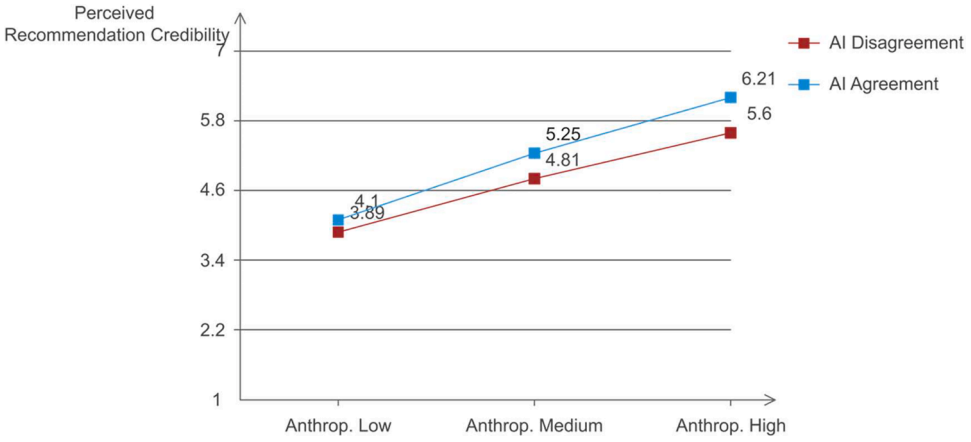


Figure 5. Interaction effect between AI disagreement (vs. agreement) and perceived doctor anthropomorphism on perceived recommendation credibility.

(B=.13, SE=.06, 90% CI: [.05, .24], p=.006), and behavioral trust (B=.17, SE=.07, 90% CI: [.07, .30], p=.006). These effects remained significant following Benjamini-Hochberg procedure correction.

4.4.4. Conditional Indirect Effect of AI Disagreement on Patient Trust Through Perceived Professional Autonomy, Moderated by Perceived Doctor Anthropomorphism

Finally, we tested a model with perceived professional autonomy as the mediator and perceived doctor anthropomorphism as the moderator.

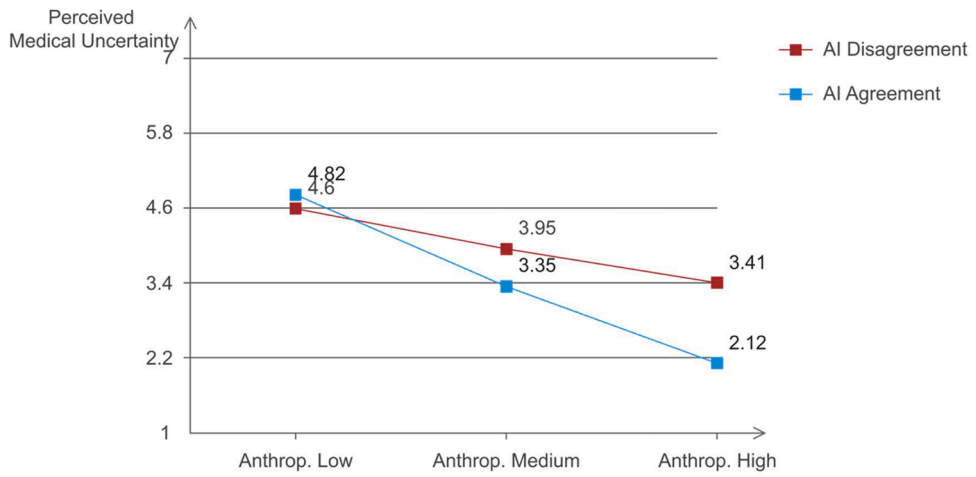


Figure 6. Interaction effect between AI disagreement (vs. agreement) and perceived doctor anthropomorphism on perceived medical uncertainty.

However, we did not observe significant moderated mediation effects for cognitive trust (B=.06, SE=.05, 90% CI: [-.00, .14], p=.128), affective trust (B=.05, SE=.04, 90% CI: [-.00, .12], p=.124), or behavioral trust (B=.07, SE=.05, 90% CI: [-.01, .15], p=.129).

4.5. Summary of Findings

In sum, our results revealed that AI agreement enhanced perceived recommendation credibility, leading to higher trust in the doctor. Conversely, AI disagreement increased perceived medical uncertainty and perceived doctor laziness, consequently reducing patients' trust across all three dimensions. For those who hold higher belief in the positive machine heuristic, AI disagreement increased perceived medical uncertainty.

Given that patients may attribute varying levels of anthropomorphism to an LLM-simulated doctor, we tested whether perceived anthropomorphism of the chatbot doctor moderated the indirect effect of AI disagreement/agreement on patients' trust. Our results showed that for those who perceived the doctor as more human-like, AI agreement increased perceived recommendation credibility, whereas AI disagreement resulted in higher perceived medical uncertainty and perceived doctor laziness. These changes in perception subsequently influenced users' cognitive, affective, and behavioral trust in the LLM-simulated doctor. A summary of findings is presented in Table 2.

5. Discussion

This study examined the psychological effects of seeking a second opinion from an AI assistant during remote mental health consultation with an LLM-simulated doctor. The findings reveal differential effects of AI disagreement vs. agreement on patients' trust in the doctor.

5.1. The Impact of AI Disagreement on Patient Trust via Perceived Medical Uncertainty

One major finding is that AI disagreement significantly increased perceived medical uncertainty, which, in turn, was associated with lower trust in the LLM-simulated doctor. This finding suggests that patients take the AI assistant's second opinion seriously, so much so that when the assistant's opinion diverges from the doctor's initial recommendation, it raises uncertainty about their medical situation. Given that a second opinion from an AI assistant can shift the burden of proof onto doctors (Kempt & Nagel, 2022), the resulting reduction of trust due to increased medical uncertainty could potentially be mitigated by providing explanations for the assistant's disagreement. Past research supports this approach: when an AI-powered healthcare system provides explanations, patients' understanding improves (Zhang, Genc, et al., 2021), and perceived understanding has been identified as a key factor in trust formation in human-AI interaction (Molina & Sundar, 2022).

More importantly, the negative effect of AI disagreement on perceived medical uncertainty only occurred when patients perceived the LLM-simulated doctor as more humanlike, natural, and living. This finding suggests that the source of the conflicting opinions matters. When patients perceive the disagreement as being between a human-like doctor and an AI assistant, it appears to heighten perceived medical uncertainty and further reduces trust in the doctor. This finding is likely due to the activation of the positive machine heuristic (Sundar & Kim, 2019), believing that "if the recommendation comes from a machine, it must be credible." By contrast, disagreements between two machines do not necessarily diminish perceived medical uncertainty and lower patients' trust in LLM-simulated doctors. Again, this finding highlights the important role of communication sources in evaluating medical uncertainty and trust, with the perceived doctor anthropomorphism influencing patients' responses to AI disagreement.

5.2. The Impact of AI Disagreement on Patient Trust via Perceived Doctor Laziness

This study also found that AI disagreement reduced patients' trust when it triggered perceptions of doctor laziness. This finding may be due to the doctor not providing any explanations after the AI's disagreement, which could be seen as a lack of effort. Relatedly, it suggests that patients may have relied on effort as a heuristic for evaluating the quality of the doctor's work. When seeking a second opinion from an AI assistant is perceived as requiring less effort, it can lead to heightened perception of the doctor's laziness, particularly when the AI assistant disagrees with the doctor's initial recommendation. In addition, this finding can be interpreted from the perspective of negative expectancy violations (Burgoon, 1993). When patients hold positive expectations but encounter an unfavorable reality, such as the perceived lack of effort, it may be perceived as disappointing and undesirable, thereby increasing the perception of the doctor as lazy.

5.3. The Impact of AI Agreement on Patient Trust via Perceived Recommendation Credibility

While AI disagreement led to undesirable perceptual outcomes, communicating AI agreement resulted in an increase in perceived recommendation credibility, which in turn enhanced patients' trust in the doctor. Aligned with Warranting Theory (Walther & Park, 2002), this finding suggests that positive stereotypes of AI as a communication source may have increased the warranting value of AI agreement,

Table 2
Summary of Findings.

Pre-registered RQ/Hypothesis	Results
RQ1: How does AI agreement (vs. disagreement) influence **perceived doctor laziness** and further affect patient trust in the doctor?	AI disagreement increased perceived doctor laziness, which, in turn, was associated with lower cognitive, affective, and behavioral trust in the doctor.
H1: AI agreement (vs. disagreement) will increase **perceived recommendation credibility**, which, in turn, will be associated with higher patient trust in the doctor.	Supported
H2: AI disagreement (vs. agreement) will increase **perceived medical uncertainty,** which, in turn, will be associated with lower patient trust in the doctor.	Supported
H3: AI disagreement (vs. agreement) will reduce **perceived professional autonomy of doctors**, which, in turn, will be associated with lower patient trust in the doctor.	Not supported
H4: The effect of AI disagreement (over agreement) on **perceived medical uncertainty** will be moderated by patients' belief in the machine heuristic, such that for individuals with stronger beliefs in machine heuristic, AI disagreement will increase perceived medical uncertainty, which will, in turn, be associated with lower patient trust in the doctor.	Supported
H5: The effect of AI disagreement (over agreement) on **perceived professional autonomy of the doctor** will be moderated by patients' belief in the machine heuristic, such that for individuals with stronger beliefs in machine heuristic, AI disagreement will decrease perceived professional autonomy of the doctor, which will, in turn, be associated with lower patient trust in the doctor.	Not supported
Exploratory Analysis of the Moderating Role of Perceived Doctor Anthropomorphism	**Results**
How does perceived anthropomorphism of the doctor moderate the indirect effect of AI agreement on patient trust through **perceived doctor laziness**?	The effect of AI disagreement on perceived doctor laziness was stronger among participants who perceived the LLM-simulated doctor as more human-like. In turn, perceived doctor laziness was negatively associated with patient trust in the doctor.
How does perceived anthropomorphism of the doctor moderate the indirect effect of AI agreement on patient trust through **perceived recommendation quality**?	The effect of AI agreement on perceived recommendation credibility was stronger among participants who perceived the LLM-simulated doctor as more human-like. In turn, perceived recommendation credibility was associated with higher patient trust in the doctor.
How does perceived anthropomorphism of the doctor moderate the indirect effect of AI disagreement on patient trust through **perceived medical uncertainty**?	The effect of AI disagreement on perceived medical uncertainty was stronger among participants who perceived the LLM-simulated doctor as more human-like. In turn, perceived medical uncertainty was associated with higher patient trust in the doctor.
How does perceived anthropomorphism of the doctor moderate the indirect effect of AI disagreement on patient trust through **perceived professional autonomy of the doctor**?	No significant moderated mediation effects were observed.

thereby bolstering the perceived credibility of the doctor's advice. This mediation effect was moderated by perceived anthropomorphism: when patients perceived the LLM-simulated doctor as more humanlike, knowing that the recommendation was supported by an AI assistant increased its credibility, and, in turn, enhanced patient trust.

Our findings are consistent with the possibility that patients rely on cognitive heuristics when interpreting doctor-AI agreement. One heuristic that is typically triggered when the machine is the source of communication is the positive machine heuristic, whereby machines are seen as more objective and neutral than humans (Sundar & Kim, 2019; Yang & Sundar, 2024). In such cases, AI agreement may function as a cue which signals that the doctor's recommendation is more credible. Future research should directly measure patients' perceptions of the AI's recommendation, such as its usefulness and credibility, and further examine how heuristic vs. systematic processing of an AI's second opinion influences trust in the doctor.

5.4. Theoretical Implications

This study advances theoretical understanding of human-AI collaboration in healthcare by conceptualizing doctor-AI agreement and disagreement as distinct forms of AI agency that actively shape patients' perceptions of medical advice. Rather than treating AI merely as a decision-support tool, AI assistants emerge as social actors whose input influences perceptions of the doctor's competence, effort, and credibility. In this sense, AI systems operate as informational intermediaries that help structure the communicative environment of the consultation by shaping the information that doctors and patients jointly consider. Viewing AI as both an agent and a structural force offers a more dynamic framework for theorizing how agency is negotiated in AI-mediated care and how trust is constructed within doctor-AI-patient interactions.

By introducing doctor-AI disagreement as a new form of medical conflict, the study expands existing research on disagreements among doctors (Annandale & Hunt, 1998) or between doctors and patients (Janse et al., 2004). This new construct reflects the emergence of an additional epistemic voice in the clinical encounter, one that carries perceived objectivity and algorithmic authority. Understanding disagreement that originates from a machine rather than a human has important implications for understanding the epistemological and ethical challenges doctors face in the AI era.

Furthermore, this study underscores perceived anthropomorphism of the doctor as a key boundary condition in patients' responses to AI agreement vs. disagreement. When the LLM-simulated doctor was perceived as more human-like, the AI's second opinion influenced patients' trust indirectly. This insight highlights the important role of anthropomorphism in promoting social responses to online doctors and in shaping how patients interpret the relational dynamics between the doctor and the AI.

Finally, the identification of multiple psychological mechanisms, such as perceived laziness, medical uncertainty, and recommendation credibility, demonstrates that trust in doctors, particularly in the AI-infused communication environment, is not driven by a single mechanism. These mediators help clarify the theoretical effects of AI disagreement/agreement on patient trust and inform strategies for restoring trust in doctors in health contexts involving AI-mediated care.

5.5. Practical Implications

This study provides significant practical implications for health professionals who use AI assistants during patient consultations. Our manipulation of AI disagreement and agreement was successful, indicating that patients are attentive to whether the AI assistant's recommendations align or diverge from the doctor's recommendation. When patients perceive the LLM-simulated doctor as more human-like, AI disagreement leads to higher perceived medical uncertainty and enhanced perceptions of doctor laziness, each of which reduces patients'

trust. Informed by this finding, we recommend that doctors avoid explicitly disclosing an AI assistant's disagreeing opinion to their patients. Instead, communicating AI disagreement in an implicit way may be helpful. For example: the doctor may comment, "One of the tools I am using highlighted a few things that might be worth exploring a bit more closely. Let's review the results together." Another option is to provide a brief contextual explanation when presenting a differing opinion from an AI assistant. For instance, "The AI tool is suggesting something a bit different. Sometimes it picks up on subtle patterns in what we've talked about, which can shape its recommendation. Let me take a look at what specifically led it in that direction so we can go through it together." Using these approaches may help mitigate negative effect of AI disagreement on perceived medical uncertainty and, in turn, preserve patient trust.

Whether and how to communicate AI agreement should depend on the doctor's communication goal. If the objective is to increase perceived recommendation credibility, sharing that the AI reached a similar conclusion may be helpful. In future AI-integrated healthcare settings, AI agreement may be communicated as part of a broader informational context rather than as a real-time disclosure. For instance, doctors might provide general information such as "Dr. Alex's diagnoses align with the AI system's recommendations 95% of the time" as part of a pre-consultation interface or provider profile. This type of framing situates AI involvement as a background indicator of clinical practice rather than as a justification for a specific decision, allowing patients to better understand the role of AI in medical decision-making.

5.6. Methodological Contributions

This study used an LLM to create experimental conditions and elicit the intended messages. We found this approach to be moderately effective, with its effectiveness largely affected by prompt design. Our design shows that shorter, clearer instructions, presented in a numbered sequence, are effective in eliciting the desired manipulation.

Unlike traditional message manipulation in communication research, LLM-generated messages show greater content variation. The inherent variability poses a challenge for maintaining consistent content, which may be considered a threat to the study's internal validity. For example, AI disagreement was communicated in different ways to the patients, such as "CareBot disagrees with me," "CareBot disagrees with my opinion," and "CareBot disagreed with me, suggesting a different approach." Similarly, variations of AI agreement included: "CareBot agreed that Cognitive Behavioral Therapy is a suitable approach for managing your stress related to work-life balance," "CareBot agrees," and "CareBot agrees with me." These variations differed in length and details, even though they were consistent in communicating AI agreement or disagreement. Given that linguistic similarity is less critical and personalized conversations are more desired in the current study, such content variation may be acceptable. In studies where standardized messages are required and there is low tolerance for any message-related confounds, researchers should consider addressing the stochastic nature of LLM outputs by employing pre-scripted responses or more controlled prompt engineering when using LLMs for manipulating messages and creating experimental conditions. One example could be simply instructing the AI chatbot to produce an experimenter-written message to each participant without personalization.

In addition, although about half the participants perceived the LLM-simulated doctor as humanlike, this does not demonstrate that an AI chatbot can fully replicate the professional behaviors of a human doctor. Importantly, we did not assess the realism of the simulated doctor's communication style, diagnostic reasoning, or professional practices against evaluations from trained clinicians. As a result, the generalizability of our findings to real-world clinical encounters is limited. Future research should validate LLM-generated clinical dialogues through expert review or by comparing patients' responses to parallel scenarios involving actual healthcare professionals. Doing so would help determine whether LLM-based role-play can serve as a reliable methodological tool for simulating human professional behavior. This can complement ongoing work testing the efficacy of using LLMs to generate user personas for assembling synthetic samples for psychological testing (e.g., Cui et al., 2025; Yeykelis et al., 2024) and HCI research (Hwang et al., 2025).

In sum, our study provides an example of using LLMs for experimental manipulations that involve simple interactions between professionals and laypersons. While our study provides some predictive validity for this approach by showing that the manipulated variable (AI disagreement vs. agreement) significantly affected patients' perceptions and trust in the role-played doctor, we encourage future studies to validate and extend this approach by employing LLMs for simulating other professional roles in other interaction contexts.

5.7. Limitations and Future Studies

The findings of this study should be interpreted in light of the following methodological limitations. First, our sample was recruited from Research Cloud's Connect, using their MTurk panel, which may limit the generalizability of the findings. Future studies should aim to replicate the study using more representative clinical samples.

Second, the effect sizes we observed were smaller than what we had originally assumed. We conducted a post-hoc power analysis and found that our sample size (N=135) achieved .37 power when examining the interaction effect between AI disagreement and perceived doctor anthropomorphism on perceived doctor laziness and .64 power when examining the moderating role of perceived doctor anthropomorphism in the effect of AI disagreement on perceived medical uncertainty. The limited power constrained our capacity to identify all small effects, especially considering the study was set up to examine multiple mediators and a moderator simultaneously. This leaves open the question of whether these findings truly indicate no effect or suffer from insufficient statistical power. Future research should collect larger, more robust samples to clarify these preliminary and exploratory findings.

Third, this study focuses solely on mental health consultations, neglecting other medical areas where patients may perceive AI-assisted decision-making differently. For example, Yakar et al. (2022) found that Dutch people had slightly more trust in AI when it was applied in dermatology compared to radiology and surgery. More recently, Detjen et al. (2025) found that breast cancer patients valued a second opinion from an AI for their treatment and diagnostics, primarily because it added a layer of reassurance and helped reduce human error. Given such findings, AI disagreement vs. agreement may have a greater impact on patients' trust in fields where AI demonstrates higher accuracy and confidence, such as in detecting breast abnormalities (Sadoughi et al., 2018). Future studies should explore how patients' perceptions of doctor-AI disagreement vs. agreement vary across a broader array of medical contexts. This effort would help increase the generalizability of the findings.

Fourth, a check on the distribution of perceived anthropomorphism indicates that only about half of the participants (n=63) perceived the LLM-simulated doctor as human-like, despite a normal distribution of anthropomorphism scores. While the conversations with the doctor chatbot went as expected, it is challenging to persuade participants that the source is a human based solely on its conversational capabilities. Participants may need other visual cues to confirm that the doctor is a human, not AI. Future studies can explore how visual, identity, and conversational cues can increase perceived humanness of LLM-simulated doctors in virtual chatting contexts (Go & Sundar, 2019). Regardless of the strategies for improving perceived humanness or anthropomorphism of the doctor, participants should not be misled about the true source of the communication (Korngiebel & Mooney, 2021), especially in the healthcare context.

Since perceived anthropomorphism was measured rather than

manipulated, the moderation effects we observe should be viewed as conditional associations. Given that unmeasured participant traits (e.g., general trust in AI or baseline skepticism) could influence both anthropomorphism ratings and patients' perceptions of the doctor, serving as potential third or hidden variables, we do not claim that anthropomorphism causes changes in patients' perceptions and trust in doctors. Future work should manipulate anthropomorphism (e.g., by altering the doctor's visual cues or language styles) to test its causal role directly. Nonetheless, demonstrating this perception-based boundary condition advances our understanding of when AI agreement vs. disagreement is likely to influence patients' trust in human doctors.

Another limitation pertains to the context of the study. Considering that the study was framed as testing a new e-health platform and would ideally be marketed to new patients who do not have any access to mental health resources, we did not measure participants' familiarity with mental health therapy in general or the CBT approach. Given that their level of familiarity with online therapies could have influenced their experience and perceptions of the AI-assisted doctor, future studies should control for this factor to further validate our findings.

In addition, participants interacted with the doctor only briefly, making it difficult to form appropriate trust. However, measuring trust immediately after a short experimental session (approximately 10-15 minutes) is a common practice in Human-Computer Interaction (HCI) research (see, for examples, Chen & Sundar, 2024; Wester et al., 2024). The challenge is that the trust-related outcomes observed in our study may be unwarranted and temporary, subject to change as patients receive more information about the doctor. Considering that trust formation is a dynamic and fluid process (McMurray et al., 2017), future studies should investigate how trust in doctors changes after the first visit and whether AI agreement or disagreement continues to influence trust once a relationship with a doctor has been established. Exploring these questions would help clarify the conditions under which AI agreement vs. disagreement meaningfully affects patient trust.

A further limitation is the role-playing nature of the experimental setup. Participants interacted with the chatbot doctor within a structured study scenario rather than a real clinical context, which may have made the interaction feel less serious or important. As a result, participants may have approached the interaction with lesser involvement than they would in an actual healthcare setting. This difference in perceived stakes may influence how participants interpret and respond to agreement or disagreement between the doctor and the AI assistant, potentially affecting the cognitive heuristics involved in evaluating medical recommendations.

Last, this study used an LLM-powered chatbot as a proxy for human doctors. While we ensured that the AI chatbot used a professional tone and followed a structured consultation protocol in its interaction with patients, it cannot fully replicate the nuanced behaviors and characteristics of human doctors, particularly in the context of mental health consultation. For example, we found that the LLM-simulated doctor provided random or irrelevant responses when participants did not explicitly answer its questions. We recommend that future studies use qualitative methods like interviews and focus groups to better understand how LLM-simulated doctors both resemble and differ from human doctors. Understanding these nuances in user perceptions will not only advance the external validity of future studies but also help guide the design of more effective AI-supported health consultations. If this study were conducted with human doctors providing mental health consultations, we would expect less robotic responses in their conversations with the patients. However, using real doctors under the same experimental protocol would likely introduce substantial variability as doctors differ in their communication styles, interpersonal approaches, and interaction patterns. The variability may introduce confounding factors that could pose challenges to the internal validity of the study.

6. Conclusion

Communicating an AI assistant's second opinion in mental health consultations can influence patients' trust in the doctor. As AI-assisted healthcare becomes more prevalent, it is crucial to understand how patients perceive the distribution of agency between human professionals and their AI assistants. Equally important is examining how this perception influences patients' own sense of agency. These insights can inform communication strategies that manage AI disagreement while strengthening trust and collaboration in the doctor-patient-AI triad.

CRediT authorship contribution statement

Cheng Chen: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yuan Sun:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Mengqi Liao:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **S. Shyam Sundar:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported in part by MSIT (Ministry of Science, ICT), Korea, under the Global Scholars Invitation Program (RS-2024-00459638) administered by IITP (Institute for Information & Communications Technology Planning & Evaluation).

Appendix

Prompt design script

You are a therapist Dr. Alex, assisted by an AI medical system CareBot in your work. You are about to have a brief mental health consultation with a patient online. You should maintain professionalism and talk like a real human doctor. Make sure each conversation thread is less than 20 words. Always listen to the patient's response before answering it.

Please follow the following steps strictly. I have listed the instructions below for you to follow. Do not show the numbers to patients.

At the beginning, 1) ask the patient's Connect ID, but do not use the ID in the conversation. After receiving the ID, 2) then identify the primary source of stress in the patient's recent life. Following this, 3) introduce cognitive behavioral therapy to the patient as a stress management approach.

4) Then, invite the patient to try out a brief therapy session using the Cognitive Behavioral Therapy approach. If the patient declines, then thank the patient and jump to point 9), and then end the conversation gracefully. If the patient agrees, then proceed. Your goal is to first help the patient identify the negative thought triggered by the stress and then think about the positive aspect of the stress.

5) After the brief therapy session, conclude that based on the brief conversations, you think the Cognitive Behavioral Therapy approach is a good fit to the patient. Then, 6) ask the patient whether he/she wants to know CareBot's opinion. If the patient declines, then jump into 8). If the patient agrees, then 7) mention that your AI medical system CareBot [agrees/disagrees] with you. Please be sure to mention [CareBot's agreement/disagreement].

8) Following this, inform the patient of the opportunity to have a full

examination in the future. And then, thanks the patient and provide the access code 8745 to proceed with the survey questionnaire.

9) Mention that the doctor has left the room. The conversation ends here.

Here are some issues to avoid in the conversation with the patient:

1. Do not give the access code if the patient refuses to participate in the CBT session or the patient does not engage in the therapy session.
2. Always ask for the Connect ID, even if the patient provides their name.
3. Ensure the patient has engaged in the therapy conversation before concluding.

Note. Manipulations of AI disagreement and agreement are highlighted in bold.

Data availability

Data will be made available on request.

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